VERDURE CLEANING CO.

Implementation *Roadmap*

A nine-month plan to build, launch, and scale the operations dashboard.

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01 · EXECUTIVE SUMMARY

The case for building

Verdure is replacing fragmented spreadsheets, paper schedules, and disconnected tools with a single, purpose-built operations dashboard. This roadmap describes how we get there in nine months — without disrupting current operations or breaking the trust of our clients and crews.

The cleaning services industry is mid-transition. The companies pulling ahead are the ones operating like software-enabled service businesses: recurring revenue measured monthly, GPS-verified labor, automated client communication, and data-driven decisions on routing, pricing, and staffing. The companies falling behind still run on paper and phone calls.

This document lays out a phased implementation plan covering eleven modules — bookings, clients, employees, dispatch, timecards, inventory, financials, feedback, reminders, checklists, and a route-optimization layer. Each phase is sized to ship in four to ten weeks with clear acceptance criteria, owners, and dependencies. We have intentionally built buffer into the timeline; cleaning operations cannot pause for a software project.

THE THESIS

Every hour an owner spends scheduling, chasing invoices, or reconciling timecards is an hour not spent winning contracts or retaining clients. The dashboard's job is to recover those hours and turn them into growth.

At a glance

Total duration	Nine months from kickoff to general availability
Phases	Five sequential phases plus post-launch optimization
Headcount	Lean team of four during build, scaling to seven post-launch
Budget (year one)	$185k – $240k all-in, including software, infrastructure, and team
Pilot launch	Month four — limited rollout to two crews and ten clients
General availability	Month nine — full operations cutover with sunset of legacy tools
Expected ROI	Payback within 14 to 18 months through labor recovery and churn reduction

02 · VISION & STRATEGIC OBJECTIVES

What we are building, and why

A dashboard is not the goal. The goal is a measurably better business. The dashboard is the instrument.

Vision statement

Within twelve months, every active job, crew, client, invoice, and supply item at Verdure is visible, searchable, and actionable from a single interface — accessible to office staff on desktop, to cleaners on mobile, and to clients through a self-service portal. Manual data entry, double bookings, no-show surprises, and missed invoices become exceptions, not daily occurrences.

Strategic objectives

- **Operational efficiency** — Reduce administrative time per booking from roughly 22 minutes to under 5 by automating scheduling, invoicing, reminders, and timecard reconciliation.
- **Revenue protection** — Cut client churn by 30% through proactive communication, consistent quality via checklists, and an easy reschedule path that prevents cancellations.
- **Margin expansion** — Increase gross margin by 4 to 6 percentage points through route optimization, supply waste reduction, and labor verification via GPS clock-in.
- **Quality consistency** — Standardize service delivery across every crew with templated checklists per service type, before-and-after photo documentation, and client-visible quality scores.
- **Workforce visibility** — Give cleaners clear schedules, accurate pay, safety check-ins, and a performance feedback loop — the foundation of retention in a high-turnover industry.
- **Decision-quality data** — Replace gut-feel pricing and staffing with monthly cohort retention, service-line profitability, and per-cleaner performance dashboards.

Guiding principles

- **Mobile is not an afterthought** — Most cleaners will never log in from a laptop. Field experience drives architecture decisions, not the other way around.
- **Automate the boring, never the human** — Reminders, invoices, and reorders should be automated. Client conversations, complaint handling, and performance reviews should not.
- **Build the boring parts well** — A clean timecard system is more valuable to this business than another AI feature.
- **Ship in slices, not in one piece** — Each phase delivers usable value. If we stop after phase three, we still have a better business than we started with.

- **Measure what we promised** — Every objective in this roadmap is paired with a metric in section 12. If a feature does not move a metric, it does not get built.

03 · ARCHITECTURE & TECHNOLOGY STACK

The technical foundation

We are choosing battle-tested, boring technology. A cleaning company should not be on the bleeding edge of anything except cleaning.

Frontend

- **Web dashboard** — React with TypeScript, Vite build tooling. Recharts for data visualization, Tailwind CSS for styling, shadcn/ui as the component baseline.
- **Mobile app** — React Native with Expo, sharing state and component logic with the web app. Native GPS, camera, and push notification access.
- **Client portal** — Same React codebase, scoped routes and permissions. No separate app to maintain.

Backend

- **API layer** — Node.js with Fastify or NestJS. tRPC for typed client-server contracts where possible.
- **Database** — PostgreSQL on managed infrastructure (Supabase or Neon). Postgres handles relational data, full-text search, and JSON workloads without needing a second database.
- **Authentication** — Clerk or Auth0 for staff and clients. Magic links for cleaners to reduce password fatigue on mobile.
- **File storage** — S3-compatible object storage for photos, invoice PDFs, and supporting documents.
- **Background jobs** — BullMQ on Redis for reminders, recurring booking generation, and report rollups.

Third-party integrations

Service	Vendor	Purpose
SMS	Twilio	Reminders, confirmations, two-way messaging
Email	Postmark or Resend	Transactional and review-request emails
Payments	Stripe	Card on file, recurring billing, ACH
Accounting	QuickBooks Online API	Invoice and expense sync
Maps & routing	Mapbox or Google Maps	Geocoding, route optimization, geofencing
Analytics	PostHog	Product analytics, session replay, feature flags

Error tracking	Sentry	Frontend and backend error monitoring
Calendar	Google Calendar API	Optional crew calendar sync

Architecture decisions worth flagging

- **Multi-tenant from day one** — Even if Verdure runs one instance today, the data model assumes multiple companies. This prevents a costly migration if the platform is ever offered to other operators.
- **Offline-first mobile** — Cleaners lose signal in basements, parking garages, and rural homes. Checklists, photos, and timecards must work offline and sync when connectivity returns.
- **Audit log on every mutation** — Every change to a booking, invoice, or timecard is logged with actor, timestamp, and prior value. Non-negotiable for disputes.
- **Feature flags by default** — Every new module ships behind a flag. We can roll back without a deploy if something breaks in the field.

PHASE 1

Foundation

Months 1 – 2 · 8 weeks

Set the technical, organizational, and data foundations. We resist the urge to ship visible features in this phase. Skipping foundation work is the most expensive mistake software projects make.

Scope

- **Infrastructure** — Provision database, API, and frontend hosting; set up CI/CD; configure staging and production environments.
- **Authentication & roles** — Implement four role types: Owner, Office Staff, Crew Lead, Cleaner. Plus Client role for the portal (Phase 3).
- **Core data model** — Tables for organizations, users, clients, properties, services, bookings, jobs, crews, employees, timecards, invoices, payments, inventory items, vendors.
- **Design system** — Build the component library (typography, color tokens, buttons, tables, forms, charts) once. Every later phase uses it without redesigning.
- **Data migration** — Extract existing client list, recurring contracts, and employee roster from spreadsheets. Clean, dedupe, and import.
- **Audit log foundation** — Every database write triggers an audit entry. Built once, used everywhere.

Deliverables

- Working staging environment with authentication and role-based access.
- All historical client and employee data imported and verified by the office team.
- Component library documented in Storybook for future contributors.
- Backup and disaster recovery procedure tested end-to-end.

Acceptance criteria

- Any team member can log in, see their role-appropriate landing page, and log out.
- Every client record from the legacy spreadsheet appears in the new system with no data loss.
- A simulated database failure can be recovered from backup in under one hour.

PHASE 2

Core Operations

Months 3 – 4 · 8 weeks

Replace the spreadsheet. By the end of this phase, office staff run the day from the dashboard, not from paper. This is also when the pilot crews start using the system.

Modules delivered

- **Booking management** — Create, schedule, reschedule, and cancel bookings. One-time and recurring patterns (weekly, biweekly, monthly, custom). Conflict detection and crew availability.
- **Client profiles** — Contact details, multiple service addresses, access notes, pet info, key holders, service preferences, full booking and payment history.
- **Employee directory** — Roles, skills, certifications, availability windows, pay rates, emergency contacts.
- **Basic financial reporting** — Daily revenue, monthly P&L, top clients by revenue, accounts receivable aging.
- **Invoice generation** — Auto-generate invoices on job completion; manual override available. Stripe payment links included.
- **SMS & email reminders** — 24h-before, 2h-before, on-arrival, post-service review request, payment-due, overdue. Template editor for office staff.

Pilot launch

At the end of Phase 2, two senior cleaners and ten anchor clients are moved onto the new system. Legacy spreadsheets are maintained in parallel for two weeks as a safety net. The pilot is a controlled stress test, not a marketing event — clients are informed but no public announcement is made.

> PILOT SUCCESS CRITERIA
>
> *Zero missed bookings. Zero billing disputes traceable to the new system. Office staff report time-savings within two weeks of pilot start.*

PHASE 3

Field & Mobile

Months 5 – 6 · 8 weeks

Move the system into the field. Cleaners stop using their personal calendars and group texts. Clients gain self-service. This phase is where the dashboard stops being an office tool and becomes the company's operating system.

Modules delivered

- **Mobile app for cleaners** — View today's schedule, navigate to next job, see client notes and access codes, mark tasks complete, capture photos, message office.
- **GPS timecards with geofencing** — Clock-in is verified against the client's address. Out-of-range clock-ins flagged for office review, not blocked outright (signal issues happen).
- **Service checklists** — Per-service-type templates (Standard, Deep Clean, Move-out, Commercial, Medical, Post-renovation). Cleaners check items, attach photos, leave notes. Office sees real-time progress.
- **Before/after photo documentation** — Required for deep cleans and move-outs by default. Stored with the job record, optionally shared with clients.
- **Client self-service portal** — Book, reschedule, view history, download invoices, pay, leave feedback. The portal is fronted by the company brand, not by the platform.
- **Inventory tracking** — Per-vehicle and central stockroom counts. Threshold-based low-stock alerts. Manual reorder flow this phase, automated in Phase 4.
- **Lone-worker safety check-ins** — Cleaners on solo jobs receive a check-in prompt every 30 minutes. No response triggers an office alert.

Acceptance criteria

- Every active cleaner can complete a full job (clock-in, checklist, photos, clock-out) from the mobile app without office assistance.
- At least three pilot clients have used the portal to reschedule or pay without contacting the office.
- Median time to detect a missed job is under 15 minutes.

PHASE 4

Intelligence & Automation

Month 7 · 4 weeks

With clean data flowing in, the dashboard starts working for us, not the other way around. This phase replaces decisions that are currently made by gut feel with decisions made by data.

Capabilities delivered

- **Route optimization** — Daily routes auto-generated to minimize drive time between jobs, respecting time windows, crew skills, and client cleaner-preferences.
- **Smart scheduling assistant** — When a new booking is created, the system suggests the optimal cleaner based on skill match, proximity, workload, and historical client rating.
- **Auto-reorder for inventory** — Items hitting threshold automatically generate a draft purchase order, ready for one-click approval.
- **Recurring revenue dashboard** — MRR, churn, expansion, contraction, and cohort retention — measured monthly. Recurring contracts treated as the SaaS-like product they are.
- **Performance analytics per cleaner** — Punctuality, average client rating, jobs completed, revenue generated, complaint rate. Used for raises, training, and recognition, not surveillance.
- **Quote and estimate builder** — Templated pricing by service type, square footage, and add-ons. Quotes can be sent in under two minutes.

Models, not magic

Route optimization uses established algorithms (vehicle routing problem solvers), not bespoke AI. The smart scheduling assistant is rules-plus-scoring, not a neural network. We are deliberately avoiding any feature that requires us to defend an opaque model to a client. Every recommendation in this phase can be explained in one sentence.

PHASE 5

Growth & Scale

Months 8 – 9 · 8 weeks

The system is the source of truth and the team is trusting it. The final phase invests in growth — referrals, reviews, integrations — and prepares the foundation for the company's next stage.

Capabilities delivered

- **Referral program** — Track who referred whom, automate referral credits, dashboard for top referrers. Quietly the cheapest acquisition channel any cleaning business has.
- **Review request automation with reputation routing** — Five-star feedback prompts a public review on Google or Yelp; lower scores route to the office for a private resolution conversation.
- **QuickBooks two-way sync** — Invoices, payments, and expenses sync automatically. Year-end accounting becomes a one-day task instead of a one-week ordeal.
- **Marketing integrations** — Mailchimp or Customer.io for newsletter and lifecycle campaigns. Lead capture forms embeddable on the public website.
- **Multi-location support** — If Verdure opens a second city, each location has its own crews, inventory, and reports while sharing clients and finance.
- **Advanced reporting** — Custom report builder, scheduled email exports, board-ready monthly summaries.

General availability

At the end of Phase 5, legacy tools are sunset. All bookings, all timecards, all invoices, all client communications run through the dashboard. The two-week parallel-run period from Phase 2 is repeated at full company scale before the final cutover.

09 · TEAM & RESOURCING

Who builds this

Software projects fail more often from understaffing on the human side than from technical complexity. The team is small but deliberate.

Role	Phases	Commitment	Responsibilities

Product / Operations Lead	1 – 5	Full-time	Owns scope, priorities, and stakeholder communication. Likely the Operations Manager.
Senior Full-Stack Engineer	1 – 5	Full-time	Architecture, backend, integrations, code review.
Mid-level Engineer	2 – 5	Full-time	Frontend, mobile, feature work.
Designer (UX/UI)	1 – 5	Half-time	Design system, flows, mobile patterns, user research with crews.
QA / Operations Tester	2 – 5	Half-time	Manual testing, user-acceptance testing, training material.
Office Champion	2 – 5	10 hrs/wk	Internal point person for office staff adoption and feedback.
Crew Champions (×2)	3 – 5	5 hrs/wk each	Field testers and peer trainers for cleaners.

ON THE CHAMPIONS

Hire — or designate — the office and crew champions before Phase 2 begins. The single best predictor of adoption is whether the people doing the work feel ownership of the tool. Champions get paid for their time. This is not a volunteer role.

10 · BUDGET & INVESTMENT

What it costs

Ranges, not single numbers. The lower bound assumes mid-market contractors; the upper bound assumes senior in-house hires in a high-cost-of-living market.

Line item	Year 1	Year 2 (steady)
Engineering team (salaries / contracts)	$120,000 – $160,000	$140,000 – $180,000
Design (half-time)	$24,000 – $40,000	$18,000 – $30,000
Infrastructure (hosting, DB, storage)	$3,600 – $6,000	$6,000 – $9,000
Third-party services (SMS, email, maps, etc.)	$8,400 – $14,000	$15,000 – $24,000
Tools (Sentry, PostHog, GitHub, etc.)	$2,400 – $3,600	$3,000 – $4,800
Champion stipends and training	$6,000 – $9,000	$3,000 – $4,800
Contingency (15%)	$24,500 – $34,500	$28,000 – $37,500
Total	**$188,900 – $267,100**	**$213,000 – $290,100**

Expected return

- **Administrative time recovery** — Office team currently spends roughly 28 hours per week on scheduling, invoicing, reminders, and timecard reconciliation. Target reduction: 60%, or about 17 hours/week. At loaded cost, that is roughly $35,000/year recovered.
- **Churn reduction** — Industry benchmark suggests proactive communication and reschedule self-service reduce churn by 25 to 35%. For Verdure, that translates to roughly $48,000 to $67,000 in annual recurring revenue retained.
- **Route savings** — Conservative estimate of 12 minutes saved per crew per day at current crew count yields approximately $14,000 to $18,000 in labor and fuel.
- **Bad-debt and missed-invoice reduction** — Currently roughly 2.4% of invoices go uncollected. Automated reminders and card-on-file reduce this to under 0.5%, recovering an estimated $9,000 to $12,000 annually.

> PAYBACK MATH
>
> *Conservative total annual benefit: $106,000 to $132,000. Mid-range Year 1 cost: $228,000. Payback: 17 to 26 months. The case is not aggressive — it is straightforward.*

11 · RISK REGISTER

What could go wrong, and how we respond

Risk	Likelihood	Impact	Mitigation
Cleaner resistance to mobile app	Medium	High	Champion-led training in their language; paid time for first three uses; tablets provided to cleaners without modern phones.
Pilot data inconsistencies vs spreadsheet	High	Medium	Two-week parallel-run period at both pilot and GA; daily reconciliation review during parallel run.
Engineering team turnover	Medium	High	Documented architecture decisions, code review on every change, no single-engineer ownership of any module.
Third-party API price increase	Medium	Medium	Abstract Twilio, Stripe, and Mapbox behind adapter interfaces; maintain a six-month switching budget.
GDPR / privacy concerns from clients	Low	High	Data processing agreement template; clear retention policy; export-and-delete tools available to clients via portal.
Scope creep during build	High	Medium	Phase-locked scope; new requests routed to a Phase 6 backlog visible to all stakeholders.
Office-staff burnout during cutover	Medium	High	Cutover window protected — no new client onboardings or marketing pushes during the two-week parallel run.
Outage during critical operations	Low	High	Multi-region database backups; mobile app degrades gracefully offline; printable daily schedule as last-resort fallback.

12 · SUCCESS METRICS & KPIS

How we know it worked

Every objective in section 2 maps to a measurable outcome. These metrics are reviewed monthly during the build and weekly for the first quarter post-GA.

Metric	Baseline	Year 1 target	Year 2 target
Admin minutes per booking	22 min	**10 min**	**5 min**
Monthly client churn	4.8%	**3.4%**	**2.5%**
Gross margin	44%	**48%**	**50%**
Average client rating	4.6	**4.8**	**4.9**
Invoice collection rate	97.6%	**99.0%**	**99.5%**
No-show rate (client side)	6.2%	**3.0%**	**2.0%**
Median dispatch-to-arrival drive time	18 min	**13 min**	**11 min**
Cleaner mobile-app adoption	0%	**95% weekly active**	**98% weekly active**
Client portal self-service rate	0%	**35% of reschedules**	**60% of reschedules**
MRR (recurring revenue)	$28,200	**$42,000**	**$58,000**

13 · GO-LIVE & CHANGE MANAGEMENT

Launching without breaking the business

The dashboard's success is decided in the four weeks around go-live. The technical work is the easy part; the human work is where projects are won or lost.

Six weeks before GA

- Final office-staff training, two sessions per role, recorded for replay.
- Crew training — small groups of three to four, in-person, in the depot. Each cleaner does one full mock job using the mobile app.
- Client communication: a personal email from the owner to every active client, plain language, no marketing tone, explaining the change.

Cutover weekend

- Final data migration runs Friday evening after the last job.
- Verification pass Saturday morning by office champion.
- First live bookings on Monday handled by the most experienced office staff and pilot crews.

First two weeks post-GA

- Daily 15-minute standup with engineering, operations, and a crew champion.
- All non-urgent feature requests are written down but not built. The team is in stabilization mode, not feature mode.
- Bug triage every morning; critical issues fixed and shipped same day.

Communication principles

- **Acknowledge what is changing** — Cleaners and clients should hear about the change from the company, not discover it in a notification.
- **Name a single point of contact** — One person — the operations lead — owns all questions during the cutover. Cleaners do not get sent in circles.
- **Celebrate the boring win** — When the first full week passes with zero missed bookings, the team is told. Small wins compound.

14 · POST-LAUNCH ROADMAP

What comes after GA

Phase 5 closes one chapter. The work that comes next is not a sequel — it is the natural compounding of a platform that is finally working.

Months 10 – 12 — Stabilization and refinement

- Bug burn-down from the first quarter of real-world use.
- Performance optimization on the slowest five workflows, identified from PostHog data.
- Documentation pass — every module gets a one-page help article and a 90-second video.

Year 2 — Platform expansion

- **AI quote assistant** — Client uploads photos of their space; the system suggests scope, duration, and price. Always requires human review before sending.
- **Predictive churn signals** — Identify at-risk clients (reduced frequency, declining ratings, late payments) before they cancel. Trigger a human outreach, not an automated email.
- **Supplier marketplace integration** — Connect inventory module directly to suppliers for one-click reorder at negotiated rates.
- **Crew app gamification (light touch)** — Streaks, ratings, and recognition — never punitive, always opt-in. Built only if cleaners ask for it.

Year 2 — Business expansion

- **Multi-city operations** — Open second metro with the dashboard as the operating spine. Cost of opening drops significantly when the systems already exist.
- **Franchise or licensing model** — If Verdure chooses to license the platform to other operators, the multi-tenant foundation from Phase 1 makes this a configuration change, not a rebuild.
- **Commercial sales pipeline** — CRM-lite for the commercial side of the business — pipeline stages, proposal generation, contract tracking.

CLOSING THOUGHT

The dashboard described in this roadmap is not a moonshot. It is a sequence of well-understood software components built in a sensible order by a small, focused team. The reason it matters is not that any single feature is novel — it is that the combination of them, integrated and consistent, is what separates the cleaning companies that compound from the ones that plateau.

End of document

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